Probability and Statistics

Continuous Random Variables and Probability Distributions

(PART 2)

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1 Uniform distribution

2 Other continuous distribution

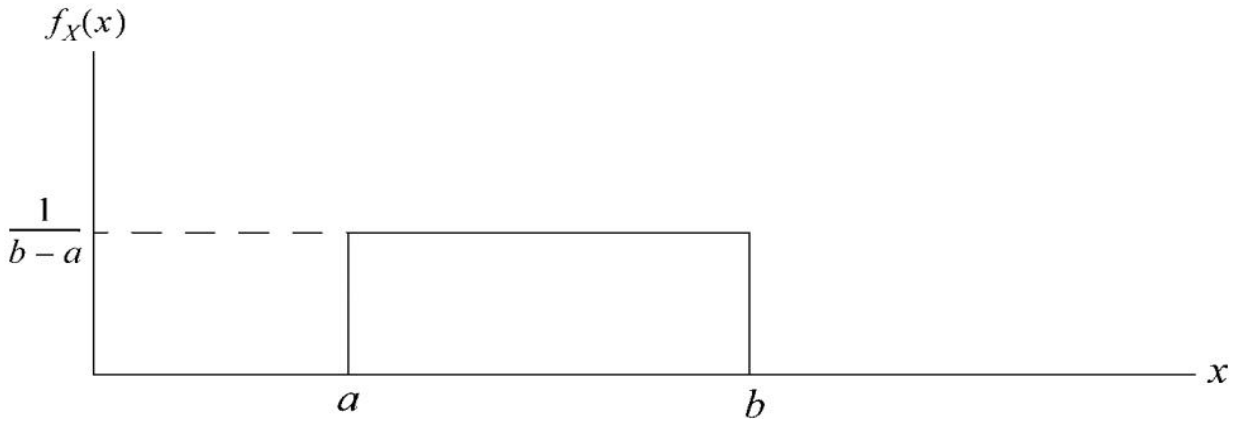
3 Probility plot

Uniform distribution

- Uniform random variable （均匀随机变量）
- PDF and CDF are of a uniform random variable are given by

$$f_X(x) = \begin{cases} \dfrac{1}{b-a} & a \le x \le b \\ 0 & \text{otherwise} \end{cases} \qquad F_X(x) = P[X \le x] = \begin{cases} 0 & x < a \\ \dfrac{x-a}{b-a} & a \le x \le b \\ 1 & x \ge b \end{cases}$$

- Used to model events that are equally likely to occur at any time within a given time interval [a, b]



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The expected value of X is given by

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)dx = \int_a^b \frac{x}{b-a}dx = \left[\frac{x^2}{2(b-a)}\right]_a^b$$

$$= \frac{b^2 - a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)}$$

$$= \frac{b+a}{2}$$

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$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x)dx = \int_a^b \frac{x^2}{b-a}dx = \left[\frac{x^3}{3(b-a)}\right]_a^b$$
$$= \frac{b^3-a^3}{3(b-a)} = \frac{(b-a)(b^2+ab+a^2)}{3(b-a)}$$
$$= \frac{b^2+ab+a^2}{3}$$

$$\sigma_X^2 = E[X^2] - (E[X])^2$$
$$= \frac{b^2+ab+a^2}{3} - \frac{b^2+2ab+a^2}{4}$$
$$= \frac{b^2-2ab+a^2}{12}$$
$$= \frac{(b-a)^2}{12}$$

Uniform distribution

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- The time a teaching assistant takes to grade a paper is a uniform random variable X between 5 minutes and 10 minutes. Find the mean and variance of the time he takes to grade a paper

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$$E[X] = \frac{10+5}{2} = 7.5$$

$$\sigma_X^2 = \frac{(10-5)^2}{12} = \frac{25}{12}$$

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Uniform distribution

- Uniform random variable （均匀随机变量）
- A random variable X is uniformly distributed between 0 and 10. Find the probability that X lies between the standard deviation and the mean

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Uniform distribution

- Uniform random variable （均匀随机变量）
- A random variable X is uniformly distributed between 0 and 10. Find the probability that X lies between the standard deviation and the mean

Since the random variable X is uniformly distributed between 0 and 10, its PDF is given by

$$f_X(x) = \begin{cases} 0.1 & 0 < x < 10 \\ 0 & otherwise \end{cases}$$

$$E[X] = \frac{10+0}{5} = 5$$

$$\sigma_X^2 = \frac{(10-0)^2}{12} = \frac{100}{12}$$

$$\sigma_X = \sqrt{\frac{100}{12}} = 2.8867$$

Thus, the probability that X lies between σ_X and $E[X]$ is given by

$$P[\sigma_X < X < E[X]] = P[2.8867 < X < 5] = \int_{2.8867}^{5} f_X(x)dx = \int_{2.8867}^{5} 0.1dx = [0.1x]_{2.8867}^{5} = 0.2113$$

1 Uniform distribution

2 Other continuous distribution

3 Probility plot

Other continuous distribution

- Weibull random variable（韦伯随机变量）

A random variable X is said to have a **Weibull distribution** with parameters α and β ($\alpha > 0, \beta > 0$) if the pdf of X is

$$f(x;\alpha,\beta) = \begin{cases} \dfrac{\alpha}{\beta^{\alpha}} x^{\alpha-1} e^{-(x/\beta)^{\alpha}} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$\mu = \beta\Gamma\left(1+\frac{1}{\alpha}\right) \quad \sigma^2 = \beta^2\left\{\Gamma\left(1+\frac{2}{\alpha}\right) - \left[\Gamma\left(1+\frac{1}{\alpha}\right)\right]^2\right\}$$

The cdf of a Weibull rv having parameters α and β is

$$F(x;\alpha,\beta) = \begin{cases} 0 & x < 0 \\ 1 - e^{-(x/\beta)^{\alpha}} & x \geq 0 \end{cases}$$

Other continuous distribution

- Lognormal random variable（对数正态随机变量）

A nonnegative rv X is said to have a **lognormal distribution** if the rv $Y = \ln(X)$ has a normal distribution. The resulting pdf of a lognormal rv when $\ln(X)$ is normally distributed with parameters μ and σ is

$$f(x;\mu,\sigma) = \begin{cases} \dfrac{1}{\sqrt{2\pi}\,\sigma x} e^{-[\ln(x)-\mu]^2/(2\sigma^2)} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$E(X) = e^{\mu+\sigma^2/2} \qquad V(X) = e^{2\mu+\sigma^2} \cdot (e^{\sigma^2} - 1)$$

Other continuous distribution

- Lognormal random variable（对数正态随机变量）
- CDF of lognormal random variable

$$F(x;\mu,\sigma) = P(X \le x) = P[\ln(X) \le \ln(x)]$$

$$= P\left(Z \le \frac{\ln(x)-\mu}{\sigma}\right) = \Phi\left(\frac{\ln(x)-\mu}{\sigma}\right) \quad x \ge 0$$

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Other continuous distribution

- Beta random variable

A random variable X is said to have a **beta distribution** with parameters α, β (both positive), A, and B if the pdf of X is

$$f(x;\alpha,\beta,A,B)=\begin{cases}\dfrac{1}{B-A}\cdot\dfrac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\cdot\Gamma(\beta)}\left(\dfrac{x-A}{B-A}\right)^{\alpha-1}\left(\dfrac{B-x}{B-A}\right)^{\beta-1} & A\le x\le B\\ 0 & \text{otherwise}\end{cases}$$

The case $A=0$, $B=1$ gives the **standard beta distribution.**

$$\mu=A+(B-A)\cdot\frac{\alpha}{\alpha+\beta}\qquad\sigma^2=\frac{(B-A)^2\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

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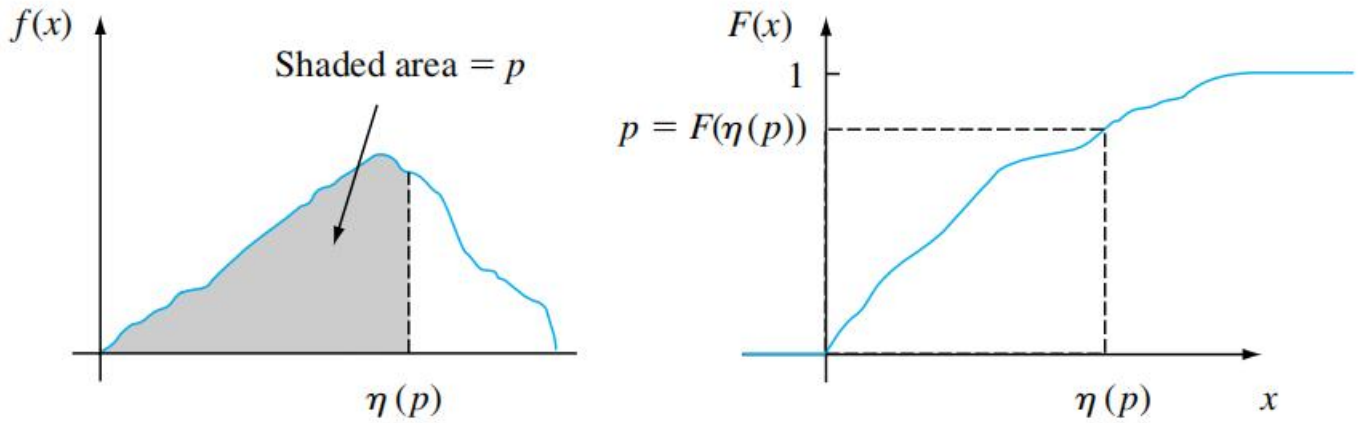
3 Probility plot

Probility plot

- Percentile of a continuous distribution （连续分布的分位点）

Let p be a number between 0 and 1. The **(100p)th percentile** of the distribution of a continuous rv X, denoted by $\eta(p)$, is defined by

$$p = F(\eta(p)) = \int_{-\infty}^{\eta(p)} f(y)\,dy$$



Probility plot

- Percentile of continuous distribution （连续分布的分位点）
- The PDF and CDF of a continuous random variable X are given by

$$f(x) = \begin{cases} \frac{3}{2}(1 - x^2) & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$F(x) = \int_0^x \frac{3}{2}(1 - y^2)\, dy = \frac{3}{2}\left(y - \frac{y^3}{3}\right)\Bigg|_{y=0}^{y=x} = \frac{3}{2}\left(x - \frac{x^3}{3}\right)$$

- What is its 50th percentile?

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Probility plot

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- What is its 50th percentile?

$$p = F(\eta(p)) = \frac{3}{2}\left[\eta(p) - \frac{(\eta(p))^3}{3}\right] \longrightarrow (\eta(p))^3 - 3\eta(p) + 2p = 0$$

Because it is 50th percentile, p = 0.5

$$\eta^3 - 3\eta + 1 = 0 \longrightarrow \eta = \eta(.5) = .347$$

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Probility plot

- Sample percentile （样本分位点）
- If we obtain *n* sample data from a process (e.g., an expriemnt) and order the *n* sample data from smallest to largest, the *i*th smallest data is $[100 * (i - 0.5)/n]$th percentile
- If the sample data set is 10, 1, 3, 6
  - 1 is $[100 * (1 - 0.5)/4] = 12$th percentile
  - 3 is $[100 * (2 - 0.5)/4] = 25$th percentile
  - 6 is $[100 * (3 - 0.5)/4] = 62$th percentile
  - 10 is $[100 * (4 - 0.5)/4] = 87$th percentile

Probility plot

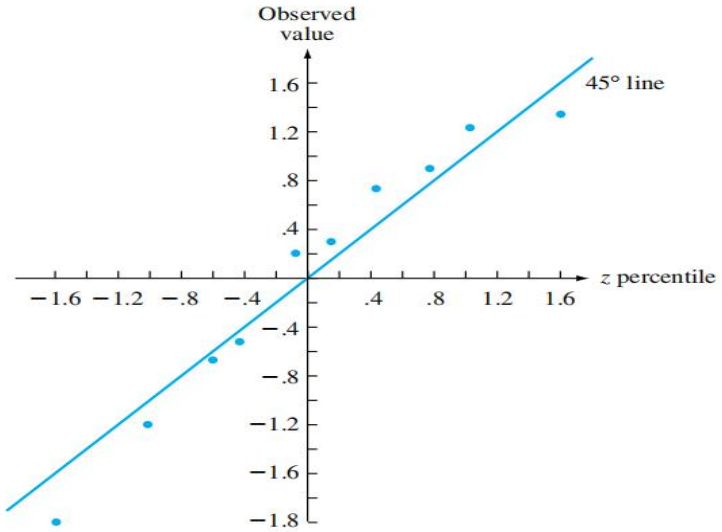
- Probility plot（概率图）
- If we expect a process follows a pecific probability distribution, the sample percentiles of sample data obtained from experiments should be close to the corresponding distribution percentiles
- Probility plot is the graph of points of ($[100 * (i - 0.5)/n]$th percentile of the distribution, ith smallest sample data)

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Probility plot

- Probility plot（概率图）
- If the sample data obtained by a process are -1.91, -1.25, -0.75, -0.53, -0.2, 0.35, 0.72, 0.87, 1.4, 1.56
- Each of them corresponds to 5th, 15th, 25th, 35th, 45th, 55th, 65th, 75th, 85th, 95th percentile
- The corresponding percentiles of standard normal distribution are -1.645, -1.037, -0.675, -0.385, -0.126, 0.126, 0.385, 0.675, 1.037, 1.645
- From the probability plot, we can know that the points are close to the line with 45° angle, meaning that the probability distribution of data from the process follow standard normal distribution

Percentage	5	15	25	35	45
z percentile	−1.645	−1.037	−.675	−.385	−.126
Sample observation	−1.91	−1.25	−.75	−.53	.20
Percentage	55	65	75	85	95
z percentile	.126	.385	.675	1.037	1.645
Sample observation	.35	.72	.87	1.40	1.56



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Home work

- Exercises of chapter 4 of text book:2, 8(b, c, d), 23, 42